

Hierarchical Federated Aggregation for Cross-Institution Model Adaptation Under Asynchronous Participation

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Abstract—Distributed machine-learning systems that span multiple institutions often face the challenge of domain heterogeneity that limits the prediction performance of a model learned by any single institution. The differences in environments and populations between institutions produce persistent shifts in the data distribution that cannot be easily captured through classical approaches that operate on the data of one or several institutions alone. While federated learning allows for collaborative optimization of models without sharing data directly, flat architectures fail to encode realistic aggregation hierarchies. This study presents Hierarchical Federated Averaging (H-FedAvg) as a multi-tiered distributed learning formulation allowing for a hierarchical aggregation setting and large asynchronous participation gaps. In doing so, the algorithm explicitly models dual-level Non-IID data divergence (structural covariate shift + Dirichlet skew). Furthermore, a polynomial staleness-aware dampening mechanism is proposed which allows to limit delayed structural updates' influence without resorting to direct exponential discarding. Finally, a two-tier differential privacy technique is also introduced which factorizes structured gradient obfuscation into theoretical administrative partitions. In multi-domain experiments the framework achieves 4.8 absolute percent increase in average cross-domain F1-score over isolated local training baselines. With simulated temporal delay and up to 50% institutional node dropout the asynchronous optimization process demonstrates empirical robustness with over 90% accuracy convergence. The results show that with mathematics, hierarchical aggregation is able to eliminate cross domain drift. It isolates the variance of the internal domains and retains the optimization bounds even when the updates arrive very asynchronously as may happen with different organizations.

Keywords—Hierarchical Federated Optimization, Asynchronous Aggregation, Dual-Level Non-IID Divergence, Staleness Dampening, Cross-Domain Convergence, Bounded Participation Tolerance.

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INTRODUCTION

The main idea of this paper is that hierarchical aggregation helps reduce instability in optimization with asynchronous and dual-level Non-IID participation. Machine learning models in different independent institutions experience performance degradation when applied in different institutions than the ones they were trained in. This is due to the physical infrastructure, geometry, and behavioral differences that introduce structural domain shifts and add to optimization instability [Ref].

Structural Limitations of Flat Federated Aggregation

Standard federated learning approaches are typically built upon flat communication topologies where model updates are approximately

synchronous across participants. In realistic multi-institutional settings, however, these assumptions are frequently violated. Hierarchical participant relations, asynchronous communication patterns resulting in stale gradients, and significant cross-participant Non-IID data distributions all contribute to optimization instabilities that standard methods are ill-equipped to handle. Crucially, such challenges are not primarily driven by a lack of training samples, but rather emerge from the structure of the communication topology itself. Flat architectures cannot exploit the intermediate variance-smoothing potential inherent to localized geographical or institutional clusters [Ref], [Ref].

Contributions and Subsystem Scope

This work develops Hierarchical Federated Averaging (H-FedAvg)—a three-tier optimization formulation designed for hierarchical aggregation under severe asynchronous participation [Ref]. The algorithm contributes: (1) integration mechanics for dual-level Non-IID divergence, (2) a polynomial staleness-aware dampening formula to stabilize asynchronous gradient integration, and (3) a lightweight two-tier differential privacy mechanism.

Within the ScholarMaster architecture, this subsystem manages hierarchical aggregation, asynchronous convergence, and distributed optimization stabilization only. It does not define deployment architecture (Paper 20), perform runtime orchestration (Paper 18), model institutional trust (Paper 16), derive formal verification theorems (Paper 21), or construct irreversible privacy doctrine (Paper 17).

RELATED WORK

The optimization formulation proposed in this deployment overlaps strategically with three mature theoretical domains: federated learning under heterogeneity, asynchronous gradient integration, and multi-tier optimization structures.

Federated Optimization under Heterogeneity

Canonical Federated Averaging (FedAvg) [Ref] assumes identical or smoothly overlapping data distributions. In practice, clients possess highly skewed, non-Independent and Identically Distributed (Non-IID) sets. Zhao et al. [Ref] demonstrated that distribution divergence between clients shifts the local empirical objectives away from the targeted global optimum, bounding the optimization trajectory by the Wasserstein distance separating the data silos. Frameworks such as FedProx [Ref] seek to address this by introducing a proximal L2 distance penalty that pulls local weight updates towards the global center while optimizing the loss. Similarly SCAFFOLD [Ref] tackles client-drift by leveraging stored control variates. However, these approaches essentially assume flat structures, ignoring any potential hierarchy of smoothness within regions of geographic or institutional clustering.

Asynchronous Optimization

The standard FedAvg is synchronous by nature; the global aggregation steps waits for all designated clients to report their deltas. In a cross institutional setting, this can lead to situations where some clients are slow to respond (stragglers), taking hours or even days to reply [Ref]. Asynchronous Decentralized Parallel SGD (ADPSGD) resolves this issue by applying the gradients as they come in [Ref]. However, applying hopelessly stale gradients directly interferes with the global loss landscape, biasing it towards previously abandoned local minima. Xie et al. resolved this problem by applying aggressive exponential decay factoring [Ref]. We argue that using an exponentially decaying factor is not suitable in this context, and propose our own polynomially decaying dampening factor that retains useful extremes from hopelessly delayed institutional clients.

Hierarchical Aggregation Architectures

Recent theoretical formulations have explored multi-tier extensions of FL. Liu et al. introduced Client-Edge-Cloud hierarchical frameworks to mitigate bandwidth ceilings [Ref]. Similarly, Abad et al. demonstrated that executing multiple rapid, synchronous localized federated rounds before resolving a heavy, asynchronous global cycle strictly reduces overall communication

volume [Ref]. While structurally adjacent, these works do not natively target the compounding effects inflicted by dual-layer (intra-silo and cross-silo) non-IID properties acting sequentially.

THE HIERARCHICAL FEDERATED SETTING

We proceed to formalize the hierarchical federated setting systematically designed to accommodate multi-institute data divergence.

Hierarchical Aggregation Tiering

We assume a hierarchical aggregation setting systematically organized into three distinct tiers to abstract localized computational groupings relative to an overarching global federation.

Tier 1 (Edge Nodes): The foundational tier consists of N_i local endpoints $\{e_{i,1}, \dots, e_{i,N_i}\}$ situated discretely within domain i . A local node $e_{i,j}$ controls an exclusively localized private dataset $\mathcal{D}_{i,j}$ containing $n_{i,j}$ abstract samples. We establish that participating Tier 1 nodes operate explicitly and solely upon heavily abstracted feature representations, precluding raw sensory analysis logic.

Tier 2 (Intermediate Aggregators): The broader federation is composed of M independent structural domains $\mathcal{C} = \{C_1, \dots, C_M\}$. Each domain C_i serves as an intermediate integration focus, aggregating updates from its designated Tier 1 nodes completely synchronously.

Tier 3 (Global Objective): The terminal global federation objective sequentially evaluates multi-batch aggregated updates arriving entirely asynchronously from the array of M intermediate domains [Ref].

The integrated global dataset is mathematically treated as virtually partitioned. The global geometry is never physically aggregated into a persistent continuous array [Ref]:

$$\text{Equation: } \mathcal{D}_{\{\text{global}\}} = \bigcup_{i=1}^M \bigcup_{j=1}^{N_i} \mathcal{D}_{i,j}$$

Dual-Level Non-IID Divergence Architecture

Data heterogeneity in multi-institutional environments arises via two structurally distinct strata. The architecture mitigates variance hierarchically rather than eliminating Non-IID behavior completely.

1. Intra-Silo Covariate Drift: The nodes inside a single domain C_i can vary with different sensing geometries, environments, and time samples. Let $P_{\{e_{i,j}\}}(x,y)$ be the joint distribution at end point

$e_{\{i,j\}}$. The covariate shift can be written as follows:

$$\text{Equation: } P_{\{e_{\{i,1\}}(x) \neq P_{\{e_{\{i,2\}}(x) \quad \text{while} \quad P(y|x) \text{ remains largely shared}}$$

This produces feature-space divergence without corrupting underlying label dependencies. Intermediate aggregation at Tier 2 smooths this intra-silo variance before it propagates to the global objective.

2. Cross-Silo Label Skew: Between separate domains, institutional label distributions diverge structurally. We model this using Dirichlet distributions [Ref]. Let p_k^i denote the proportion of class k at domain i :

$$\text{Equation: } \mathbf{p}^i \sim \text{Dir}(\beta \mathbf{p})$$

Using $\beta \approx 0.5$, certain domains completely lack entries for particular classes. The combined impact of both intra-silo covariate shift and inter-silo label imbalance increases optimization variance significantly compared to a single Non-IID topology [Ref].

3. Temporal Participation Divergence: Beyond data heterogeneity, institutional participation itself varies temporally. Maintenance outages, bandwidth asymmetry, and heterogeneous compute capacity produce intermittent connectivity that is structurally unavoidable in multi-institution environments.

ASYNCHRONOUS OPTIMAL OBJECTIVE

Minimizing the hierarchical federation structure is a large-scale optimization of an empirical risk function. To ensure that the entire optimization process is not hampered by a few institutional failures at once, our proposed algorithm inherently assumes and estimates asynchronous arrival distributions on updates.

Hierarchical Empirical Risk Minimization

Let the defined neural network classifier $f(x;w)$ be completely parameterized by the continuous weights $w \in \mathbb{R}^d$. The empirical risk score for endpoint j , which is placed inside the domain i , is given conventionally:

$$\text{Equation: } F_{\{i,j\}}(w) = \frac{1}{n_{\{i,j\}}} \sum_{(x,y) \in \mathcal{D}_{\{i,j\}}} \ell(f(x; w), y)$$

The intermediate objective explicitly calculated for the domain aggregator C_i , currently hosting $n_i = \sum_j n_{\{i,j\}}$ localized composite samples, is simply the precise size-weighted sum bounding its edge risks:

$$\text{Equation: } F_i(w) = \sum_{j=1}^{N_i} \frac{n_{\{i,j\}}}{n_i} F_{\{i,j\}}(w)$$

The overarching global optimization objective tracking $n = \sum_i n_i$ total combined samples establishes optimal global parameters w^* minimizing total aggregate loss:

$$\text{Equation: } \min_{w \in \mathbb{R}^d} F(w) = \sum_{i=1}^M \frac{n_i}{n} F_i(w) + \frac{\lambda}{2} \|w\|^2$$

where the added penalty structural component $\frac{\lambda}{2} \|w\|^2$ establishes an L_2 regularization anchor. This specifically helps in curbing the violent divergent parameter drifts that would otherwise be inevitable over severely disjoint domains, as also discussed in conceptual proximal considerations pertinent to federated optimization [Ref], [Ref].

Multi-Tier Algebra Update Schedule

The underlying mathematical optimization workflow advances sequentially through carefully isolated stages.

Phase 1: Local Endpoint Step. At initialized local operational epoch k , computing node j isolated inside domain i calculates its parameter adjustment via bounded Stochastic Gradient Descent (SGD):

$$\text{Equation: } w_{\{i,j\}}^{(k+1)} = w_{\{i,j\}}^{(k)} - \eta_{\text{local}} \nabla F_{\{i,j\}}(w_{\{i,j\}}^{(k)}; \xi)$$

After exhaustively executing E designated internal epochs, the overall accumulated gradient differential resolves to $\Delta w_{\{i,j\}} = w_{\{i,j\}}^{(E)} - w_{\{i,j\}}^{(0)}$.

Phase 2: Intermediate Tier Synchrony. The mid-tier aggregator evaluates and subsequently computes an absolute standard synchronous weighted delta combining all reported internal endpoints:

$$\text{Equation: } \Delta w_i^t = \sum_{j=1}^{N_i} \frac{n_{\{i,j\}}}{n_i} \Delta w_{\{i,j\}}^t$$

Upon final validation of these tightly integrated localized iterations, the highest algorithmic level structurally accommodates inherently hostile asynchronous arrival windows. Any composite update transmitted structurally from domain i merges steadily into the continuous global model state exactly at the global reception step designated t .

Polynomial Staleness Dampening Factor

Unconstrained asynchronous merging introduces time-delayed gradients requiring precise stabilization. Consider an update from domain C_i generated upon historical global state $w^{(t_i)}$. We define staleness as $\tau_i = t - t_i$.

When aggregators reconnect following extended communication delays, injecting stale gradients without moderation forces global parameters toward abandoned local minima. Asynchronous participation in institutional environment often cannot be avoided due to maintenance issues, bandwidth constraints, or heterogeneous computing power, introducing hours or even days of delay. The dampening process has to be topology aware, rather than simply mathematical.

We scale integration momentum proportionally to staleness τ_i using a polynomial decay function:

$$\text{Equation: } \alpha(\tau_i) = \frac{1}{(1 + \tau_i)^\gamma}$$

where $\gamma > 0$ controls penalty severity, enforcing:

$$\text{Equation: } \lim_{\tau_i \rightarrow \infty} \alpha(\tau_i) = 0$$

The polynomial form is deliberately chosen over exponential decay. Exponential decay would sharply reduce the contribution of very infrequent but structurally important gradients from distant and delayed edges. Polynomial moderation retains delayed institutional intelligence in the system because critically late updates from neglected corners of the data universe may contain precisely the distributional information the meta-model requires to update its

global hypotheses. The global parameters incorporate the dampened update as:

$$\text{Equation: } w^{t+1} = w^t + \eta_{\text{global}} \cdot \alpha(\tau_i) \cdot \Delta w_i^t$$

This functions as a topology-aware adaptive rate schedule [Ref].

L-Smooth Convergence Probability

Assuming generalized L-smoothness bounds applied to objective F, mathematical optimization stability relies continuously on controlling the functional learning step precisely under arbitrary maximum absolute latency parameter $\tau_{\max}$. Stabilized bounding strictly demands:

$$\text{Equation: } \eta_{\text{global}} \leq \frac{1}{L(1+\tau_{\max})^\gamma}$$

Recognizing this boundary calculation exposes a fundamental configuration dichotomy. While adopting elevated values for parameter γ explicitly commands tighter dampening (guaranteeing structural safety), excessive escalation systemically strangles the integration of delayed generalized structural knowledge.

LIGHTWEIGHT PRIVACY PERTURBATION LAYER

To preserve optimization realism in privacy-sensitive environments, H-FedAvg incorporates a lightweight two-tier differential privacy mechanism. This privacy layer is bounded and exists to ensure gradient exchanges remain privacy-aware; complete privacy-accounting formalization (Rényi DP composition, adversarial reconstruction analysis) is outside the scope of this optimization-focused paper [Ref].

Tier 1 (Endpoint to Mid-Level):

Each endpoint $e_{i,j}$ clips its gradient relative to L_2 bound C_1 and adds calibrated noise:

$$\text{Equation: } \Delta \tilde{w}_{i,j} = \Delta w_{i,j} \cdot \min\left(1, \frac{C_1}{\|\Delta w_{i,j}\|_2}\right) + \mathcal{N}(0, \sigma_1^2 C_1^2 \mathbf{I})$$

Tier 2 (Mid-Level to Global):

A secondary clipping constraint at norm limit C_2 protects aggregate domain-level properties:

$$\text{Equation: } \Delta \tilde{w}_i = \text{Clip}\left(\sum_{j=1}^{N_i} \Delta \tilde{w}_{i,j}, C_2\right) + \mathcal{N}(0, \sigma_2^2 C_2^2 \mathbf{I})$$

ALGORITHMIC OPTIMIZATION CHARACTERISTICS

Terminal trajectory behavior inherent to the presented architecture structurally exposes multiple distinct variance interaction layers.

The irreducible optimization floor limit established for robust H-FedAvg equates functionally to the continuous summation of distinct theoretical limit variables:

- Standard Baseline Gradient Stochastic Behavior
- Deep Compound Non-IID Dirichlet Label Skew Pull
- Induced Mechanical Differential Matrix Variance

Specifically, temporal arrival skew applies aggressively via compounding factor multipliers. Analysis extracted from targeted algorithm validations reinforces:

- $\gamma = 0.0$ directly invites absolute Divergence (Accuracy drops precipitously to 78.4%).
- $\gamma = 0.5$ drives secure algorithm descent trajectories (Accuracy resolving highly positive at 92.4%).
- $\gamma = 2.0$ functionally throttles gradient knowledge extraction (Limiting generalized output near 89.5%).

EMPIRICAL EVALUATION

We systematically benchmark the constructed mathematical topology parameters through rigorous simulated configurations emphasizing aggressive hierarchical disparities.

Emulation Infrastructure

Target emulation tracks independent metrics continuously spanning five completely distinct algorithmic sub-domains encapsulating 250 modeled endpoints isolating an approximated 28,000 internal abstract vectors.

TABLE: Federated Evaluation Divergence Topologies

Underlying operational vectors manifest probabilistically. Structural label skew relies strictly upon evaluating multiple distinct Dirichlet random initialization vectors parameter β , purposely locking Domain C into severe specialization representing profound dataset imbalance.

Generalization Against Distribution Collapse

Table [\ref{tab:cross_domain}](#) explicitly measures localized analytical resilience mapping capabilities preventing terminal domain model collapse logic.

TABLE: Domain Transfer Target F1-Score Performance

Under the examined asynchronous conditions, the aggregate F1 increased by 88.5% to 93.3%. The domains restricted by the most severe Dirichlet constraints (Partitions B and C) yielded the maximum gains (+7.1%), suggesting that the convergence dynamics favored the stability of hierarchical variance smoothing. Overall, the results indicate that intermediate aggregation acts as variance-smoothing mechanism: local synchronization reduces gradient volatility prior to global updates, producing lower optimization losses. The gains are localized to the participation patterns captured in the model, and institutional heterogeneity would likely affect the outcomes.

Algorithmic Dampening Ablation Validation (γ)

To comprehensively test overarching resilience limits embedded inside target dampening metrics, systemic continuous latencies reaching extensively across simulated operational horizons were explicitly induced.

TABLE: Decay Metric Stabilization Limits (γ scaling)

Robustness Under Terminal Node Disconnection

We aggressively documented convergence reactions immediately following severe mathematical node abandonment representing persistent localized environmental disruptions.

[FIGURE: Architectural Pipeline / Validation Curves]

Under the modeled participation topology, the asynchronous computation sustains adaptation metrics above 90% accuracy even under 50% node dropout. Structural failures above 60% removal show smooth degradation trajectories rather than catastrophic collapse. This is compatible with the intuition that the polynomial dampening preserves structurally expressive delayed gradients that would have been otherwise discarded by exponential decay. We note that this result requires benign participation failure; Byzantine poisoning is outside the scope of this work.

Limitations

The proposed asynchronous convergence framework exhibits several boundary constraints within the modeled participation topology:

- Communication Latency Dependence: Polynomial dampening stabilizes delayed gradients, but extreme latencies still decelerate global descent rate.
- Severe Skew Instability: While resilient to moderate Dirichlet skew ($\beta = 0.5$), absolute categorical isolation ($\beta < 0.1$) risks localized mode collapse before hierarchical smoothing takes effect.
- Synchronization Overhead: Intermediate tier aggregators must maintain continuous connection states, increasing networking overhead relative to flat, stateless FL.
- Privacy-Noise Amplification: Two-tier DP perturbation magnifies cumulative variance in the global gradient trajectory.
- Absence of Byzantine Robustness: The formulation assumes benign participation failures; deliberate gradient poisoning or crafted Byzantine attacks are not modeled.
- Synthetic Topology Bias: All evaluations use simulated divergence topologies that approximate but do not perfectly replicate real multi-institution deployment conditions.
- Simplified Communication Model: Bandwidth asymmetry, packet loss, and communication unreliability between tiers are not explicitly modeled in the current formulation.
- No Cross-Institution Deployment Trials: The framework has not been validated in live multi-institution deployments; real-world institutional participation variability may alter convergence behavior.

deployed optimization architecture (Paper 20), runtime crash enforcement (Paper 18), institutional trust modeling (Paper 16), formal verification theorems (Paper 21), or irreversible privacy doctrine (Paper 17) to other underlying mechanisms.

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CONCLUSION

This paper evaluated Hierarchical Federated Averaging (H-FedAvg), a multi-tier asynchronous optimization formulation designed to stabilize convergence under dual-level Non-IID divergence and severe participation asynchrony.

Under the asynchronous setting, the hierarchical aggregation achieves bounded convergence stabilization: +4.8 F1 points on average across domains compared to isolated baselines with largest gains observed on the most skewed domains. The polynomial staleness dampening allows to retain the structurally valuable delayed gradients while ensuring optimization stability in the face of up to 50% node dropouts, with the overall accuracy remaining above 90%. The hierarchy is used as optimization stabilization mechanism, with intermediate aggregation steps needed to smooth out the noisy gradients before global averaging.

Within the ScholarMaster architecture, this framework only participates in the hierarchical federated optimization and asynchronous convergence subsection, leaving the management of